Which Learner Should We Use to Estimate Heterogeneous Treatment Effects?

IS 507 Midterm Presentation

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Based on *Curth et al., *Nonparametric Estimation of Heterogeneous Treatment Effects: From Theory to Learning Algorithms.* AISTATS 2021*.

Why estimate treatment effects?

Prediction is not causation.

Same person	With treatment	Without treatment
Outcome	8 (observed)	? (unobservable)

Problem: Predicting an outcome does not tell us whether or how much a treatment changes it.

Motivation: Treatment effects vary across individuals, making accurate estimation essential for personalized decisions.

Potential impact: Identify individuals or subgroups with higher expected treatment benefits.

From ATE to CATE

Observed data: $D = \{(X_i, A_i, Y_i)\}_{i=1}^{n}$.

- X: Pre-treatment features
- $A \in \{0, 1\}$: Treatment assignment
- Y: Observed outcome

Average Treatment Effect (ATE):

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

Conditional Average Treatment Effect (CATE):

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

where $Y(1)$ and $Y(0)$ denote potential outcomes under treatment and no treatment.

Our goal: Learn $\hat{\tau}(x)$ to estimate how treatment effects vary with features.

Assumptions for CATE Identification

Three standard assumptions:

- **Consistency:** $Y = Y(A)$
- **Unconfoundedness:** $(Y(0), Y(1)) \perp A \mid X$
- **Overlap:** $0 < \pi(x) < 1$, where $\pi(x) = P(A = 1 \mid X = x)$

Under these assumptions:

$$\tau(x) = \mathbb{E}[Y \mid A = 1, X = x] - \mathbb{E}[Y \mid A = 0, X = x]$$

Our Simulation

Satisfy these assumptions by construction while varying the strength of overlap.

Two Ways to Estimate CATE

1. Plug-in Learning (S/T)

Predict outcomes under each treatment:

$$\hat{\mu}_0(x), \quad \hat{\mu}_1(x)$$

$$\boxed{\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)}$$

2. Two-step Learning (RA/PW/DR)

Estimate outcome and/or propensity models, construct pseudo-outcomes, and learn CATE through regression.

$$(X, A, Y) \xrightarrow{\text{Estimate}} \hat{\mu}_a, \ \hat{\pi}$$

$$(X, A, Y; \hat{\mu}_a, \hat{\pi}) \xrightarrow{\text{Construct}} (X, \widetilde{Y}) \xrightarrow{\text{Regression}} \hat{\tau}(x)$$

What Does the Paper Tell Us?

Main Findings

1. Learner performance varies across data-generating processes.
2. DR has favorable asymptotic properties, but simpler learners can outperform it in finite samples.
3. In the paper's simulations, DR, RA, and plug-in learners each perform best under different data-generating processes.

Key Takeaway

Learner selection should depend on the underlying data characteristics.

What Do We Still Want to Understand?

Beyond the Paper

The paper investigates different outcome structures and treatment imbalance. We want to examine their joint effects more systematically.

Our Main Research Question

How do conditional treatment overlap and the functional forms of $\mu_0(x)$ and $\mu_1(x)$ jointly affect CATE learner performance?

Experimental Factors

- Strong vs. weak conditional overlap
- Different functional forms of $\mu_0(x)$ and $\mu_1(x)$
- Sample size (secondary factor)

Proposed Data and Experimental Conditions

Property	Specification
Data	Synthetic, numerical observations
Features	25 Gaussian covariates (X)
Treatment	Binary ($A \in \{0, 1\}$)
Outcome	Continuous (Y)
Sample size	Development: 500 or 2,000; Test: 500
Overlap	Strong vs. weak conditional overlap
Outcome-generating functions	Independently vary the functional forms of $\mu_0(x)$ and $\mu_1(x)$

Why Simulation?

The true CATE $\tau(x)$ is known, allowing direct evaluation of estimation accuracy.

Training and Evaluation Plan

Training

1. Split development data into two equal folds ($K = 2$).
2. Fit nuisance models using MSE (outcome) and BCE (propensity).
3. For two-step learners, cross-fit nuisance predictions, construct pseudo-outcomes, and regress on X.

Evaluation

CATE Root Mean Squared Error (RMSE):

$$\mathrm{RMSE}_{\mathrm{CATE}} = \sqrt{\frac{1}{m} \sum_{j=1}^{m} \left(\hat{\tau}(X_j) - \tau(X_j)\right)^2}$$

- Evaluate on 500 independent test samples.
- Repeat with 10 random seeds; report mean RMSE, standard error, and runtime.

Conclusion

What We Learned

Different CATE learners exhibit different strengths across data-generating conditions.

What We Propose

Investigate how conditional treatment overlap and the functional forms of $\mu_0(x)$ and $\mu_1(x)$ jointly affect learner performance.

Expected Contribution

Provide empirical evidence on the robustness of different CATE learners.

However, simulation results may not generalize to real observational data.

References

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